

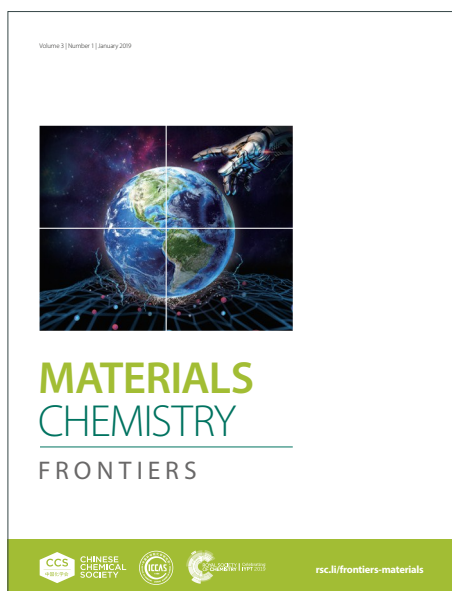
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ARTICLE

Lithium bis(oxalate)borate additive in electrolyte to improve Li-rich layered oxide cathode materials

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Lithium-rich layered oxides (LLO), as the most attractive cathode materials for high-energy lithium-ion batteries (LIBs), are plagued by poor cyclability due to structural and electrode/electrolyte interface instability. Herein, we report the synthesis of LLO and its performance enhancement by using boron-containing electrolyte additives. In a formulated 1.0 M LiPF₆ ethylene carbonate/ethyl methyl carbonate electrolyte with 0.1 M lithium bis(oxalato)borate (LiBOB), the battery assembled with Li_{1.2}Mn_{0.54}Co_{0.13}Ni_{0.13}O₂ microspheres presents a stable specific capacity of 202 mAh g⁻¹ at 0.5 C and a remarkable capacity retention of 96.4% after 100 cycles, significantly outperforming the cathode in baseline electrolyte without LiBOB. The combination of voltammetry, impedance, microscopy, spectroscopic analysis and density functional theory (DFT) calculations corroborates the beneficial effect of LiBOB on stabilizing the LLO/electrolyte interface. Reactions between LiBOB and activated oxygen radicals result in the formation of a dense cathode electrolyte interface (CEI) film (~15 nm) containing oxalate, lithium fluoride and alkyl borate species, which contributes to suppression of capacity/voltage decay of LLO. These results would provide insight in understanding the effect of boron-containing electrolyte additive on upgrading high-capacity Li-rich cathode materials.

Introduction

There is an extensive demand of advanced lithium-ion batteries (LIBs) with higher energy density for applications in portable electronics and electric vehicles.¹⁻³ Further increasing the energy density of LIBs relies largely on development of cathode materials with higher capacity. Lithium-rich layered oxides (LLO), xLi₂MnO₃·(1-x)LiMO₂ (M = Ni, Co, Mn), have attracted increasing attention as they offer high working voltage (>4.5 V vs. Li⁺/Li) and specific capacity (>250 mAh g⁻¹) compared to prevailing cathodes such as LiCoO₂ and LiFePO₄.⁴ However, LLO cathodes suffer from issues of capacity/voltage degradation, which is suggested to relate with the release of reactive oxygen species during activation of Li₂MnO₃ phase and the irreversible structure evolution upon cycling.⁵ To address these problems, extensive efforts have been made from aspects of surface and bulk modifications. For example, coating electrochemically active spinel phase or inert

components (e.g., AlF₃, Al₂O₃, etc.) is demonstrated to prevent structural damage of electrode from side reactions with electrolyte.⁶⁻⁹ Doping functional elements (e.g., Al, Sn, Nb, etc.) in LLO is proven to effectively promote lithium ion diffusion or stabilize the layer structure via enlarging lattice spacing or pillar effect.¹⁰⁻¹²

Besides, the electrolytes play a vital role in determining the performance of LLO. Substantial progress has been made in electrolyte additive chemistry and indicates that a protective film could be constructed in-situ on the cathode surface to restrain the detrimental reactions between the active electrode materials and electrolyte.¹³⁻¹⁵ Along this line, boron-containing compounds (e.g., lithium difluoro(oxalate)borate, lithium bis(oxalate)borate, etc.) have been widely studied as surface-forming additives to enhance the electrochemical stability of high-voltage cathodes such as LiNi_{0.5}Mn_{1.5}O₄ (LNMO),¹⁶⁻¹⁸ Ni-rich^{19,20} and Li-rich layered oxides.²¹⁻²⁴ Particularly, lithium bis(oxalate)borate (LiBOB) is a highly efficient additive to improve cycling property as its preferential oxidization capability. However, the mechanism of LiBOB remains elusive for LLO. Previous studies proposed that LiBOB could inhibit unwanted degeneration of Li₂MnO₃ phase,²³ while the benefit of LiBOB on Li_{1.17}Ni_{0.17}Mn_{0.5}Co_{0.17}O₂ was attributed to a LiBOB-derived CEI layer.²⁴ There is also suggestion that boron-containing additives combine with active oxygen species extracted from Li₂MnO₃ phase.^{21,22} In spite of these advances, it is desirable to further formulate boron-based electrolyte additive and straightforward elucidate the interaction between LLO and electrolytes.

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In this study, we report the effect and mechanism of LiBOB as the electrolyte additive for a high-capacity LLO material, i.e., $\text{Li}_{1.2}\text{Mn}_{0.54}\text{Co}_{0.13}\text{Ni}_{0.13}\text{O}_2$. The prepared $\text{Li}_{1.2}\text{Mn}_{0.54}\text{Co}_{0.13}\text{Ni}_{0.13}\text{O}_2$ microspheres exhibit significant improved capacity retention in carbonate electrolyte with the addition of 0.1 M LiBOB as compared to that in baseline electrolyte. Electrochemical investigations of voltammetry, differential capacity and impedance reveal the preferential oxidation of LiBOB and inhibition of solvent overoxidation during cycling. Microscopy analysis demonstrates the formation of a uniform film on the cathode surface in LiBOB-containing electrolyte, which helps to maintain structural integrity and suppresses the dissolution of transition metals. The composition of this protective layer and the interaction between LiBOB and cathode are further studied by combining spectroscopies and theoretical calculations. The results provide evidence and insight to understand the electrode/electrolyte interface chemistry of LLO cathode with boron-based electrolyte additive.

Experimental

Electrolyte and electrode preparation

Li-rich layered oxide $\text{Li}_{1.2}\text{Mn}_{0.54}\text{Co}_{0.13}\text{Ni}_{0.13}\text{O}_2$ ($0.5 \text{ Li}_2\text{MnO}_3 \cdot 0.5 \text{ LiMn}_{1/3}\text{Co}_{1/3}\text{Ni}_{1/3}\text{O}_2$) was prepared by co-precipitation method according to previous report.²⁵ Briefly, a solution of 2.0 mol L⁻¹ $\text{MnSO}_4 \cdot \text{H}_2\text{O}$, $\text{CoSO}_4 \cdot 7\text{H}_2\text{O}$ and $\text{NiSO}_4 \cdot 6\text{H}_2\text{O}$ with a molar ratio of Mn:Co:Ni = 4:1:1 was mixed with 2.0 mol L⁻¹ Na_2CO_3 aqueous solution in a stirred tank reactor for 15 h. Next, the collected precursor was washed with deionized water and dried at 80 °C under vacuum. Finally, the obtained sample was pretreated at 500 °C for 8 h in air and mixed with Li_2CO_3 . The mixture was calcined at 900 °C for 20 h in air atmosphere to obtain the target product. The prepared active material, acetylene black and polyvinylidene fluoride (PVDF) were blended with a mass ratio of 8:1:1 in N-methylpyrrolidone (NMP). The slurry was spread on the Al foil and dried at 110 °C for 12 h in vacuum. The obtained electrode sheet was cut into a disc with 10 mm diameter, and the mass loading of active material was kept at about 4.0 mg cm⁻².

LiBOB was purchased from DoDoChem and dried at 80 °C for 10 h in vacuum before use. The baseline electrolyte (denoted as baseline) was composed of 1.0 M lithium hexafluorophosphate (LiPF_6) in ethylene carbonate (EC) and ethyl methyl carbonate (EMC) (4:6 in weight). Different amount of LiBOB was dissolved in baseline electrolyte to attain the LiBOB-containing electrolytes (denoted as “n M LiBOB”, n = 0.01, 0.05, 0.1, 0.2, and 0.3). The volume of electrolyte injected to each cell was kept at 50 μL . Coin cells (CR2032) were assembled with the prepared cathode, configured electrolyte, Celgard 2500 separator and lithium metal anode. Electrolyte configuration and battery assembly were performed in an argon-filled glove box with O_2 and H_2O concentration below 0.1 ppm.

Electrochemical tests and physical characterizations

Charge-discharge tests of 2032-type cells were measured on Land and Neware battery test systems between 2.0 V and 4.8

V at 0.5 C (1 C is defined as 250 mAh g⁻¹) after two activation cycles at 0.1 C and one buffering cycle at 0.2 C under 25 °C. Electrochemical impedance spectroscopy (EIS) was tested on Ivium-n-Start workstation in a frequency range between 100 kHz and 0.1 Hz with 10 mV amplitude. Cyclic voltammetry (CV) was carried out at a scan rate of 0.1 mV s⁻¹ within 2.0–4.8 V on Ivium-n-Start workstation. Cycled batteries were disassembled in glove box and the cathodes were dried under vacuum after washing repeatedly with ultra-dry dimethyl carbonate (DMC) solvent. The crystal structure of LLO was examined by X-ray diffraction (XRD, Rigaku MinFlex600, Cu K α radiation). The morphology and elemental analysis of the cathodes before and after cycling were observed by scanning electron microscopy (SEM, JEOL JSM7500F) and transmission electron microscopy (TEM, FEI Talos F200X G2) equipped with energy dispersive spectroscopy (EDS). Inductively coupled plasma emission spectrometry (ICP, SpectroBlue) was conducted to evaluate the amount of transition metal ions (TM) deposited on cycled lithium metal anodes in different electrolytes. Fourier transform infrared spectra (FTIR, Bruker TENSORII) and X-ray photoelectron spectroscopy (XPS) were adopted to determine the surface chemical compositions of the cycled cathodes.

Theoretical calculations

All of the structures were fully optimized using the hybrid B3LYP functional^{26–30} with the basis set of 6-31G(d, p). The frequency calculations were performed at 298.15 K to verify that the stationary points were the local minima, and to obtain thermal corrections. All calculations were performed with Gaussian 16 (Revision A.03). The geometries were displayed using CYLview.

Results and discussion

Fig. 1a shows the XRD pattern and crystal structure of synthesized $\text{Li}_{1.2}\text{Mn}_{0.54}\text{Co}_{0.13}\text{Ni}_{0.13}\text{O}_2$ powder. The pattern can be indexed to a layered $\alpha\text{-NaFeO}_2$ structure (R-3m symmetry) accompanied by a distinct superlattice reflection located at 2 θ degree of 20–25°, which corresponds to the monoclinic layered Li_2MnO_3 phase (C/2m symmetry). These diffraction peaks are characteristics of Li-rich oxides. As displayed in SEM image (Fig. 1b), the LLO particles present spherical shape with an average diameter of 9 μm and an individual sphere is composed of primary particles with size of 60–70 nm.

The prepared $\text{Li}_{1.2}\text{Mn}_{0.54}\text{Co}_{0.13}\text{Ni}_{0.13}\text{O}_2$ cathode in baseline and LiBOB-containing electrolytes were comparatively tested to investigate the effect of LiBOB electrolyte additive on the performance of Li-rich oxides. A 0.1 M LiBOB additive was firstly tailored for further investigation as this composition ensures simultaneously high and stable capacity of LLO in comparison with higher or lower concentrations (Fig. S1). Fig. 1c shows the first charge-discharge curves of LLO cathodes with baseline and 0.1 M LiBOB electrolytes in the voltage range of 2.0–4.8 V at 0.1 C. The charge/discharge capacities in 0.1 M LiBOB electrolyte are 288.4/244.3 mAh g⁻¹, which are lower than those in baseline electrolyte (306.3/261.1 mAh g⁻¹). This difference indicates that LiBOB may participate in the

formation of the interface film and retard the activation of Li_2MnO_3 phase in first charging, which is consistent with the CV curves of $\text{Li}|\text{LLO}$ cells (Fig. S2). From CV curves on Al electrode (Fig. S3), the lower anodic potential and current in the electrolyte with 0.1 M LiBOB reflect the preferential oxidation of LiBOB and suppression of electrolyte decomposition. These two factors result in a decrease of capacities during the initial discharge/charge, similar to previous reports on LLO cathodes in boron-containing electrolytes.^{14,19,21,31}

Electrochemical stability of LLO was evaluated at 0.5 C for 100 cycles. As shown in Fig. 1d, a rapid capacity decay occurs in the baseline electrolyte. On the contrary, LiBOB additive endows the LLO with a superior capacity retention at about 200 mAh g^{-1} . The discharge capacity of LLO with LiBOB additive is lower in the initial 20 cycles, which could be attributed to the LiBOB-induced formation of resistant CEI layer that retards

electrode activation and traps deliverable lithium. The LLO cathode often undergoes voltage fade upon cycling, which may result from the structural distortion of the layered phase.^{14,21,22,32} The cell with 0.1 M LiBOB additive delivers a higher midpoint voltage as compared to that without additive, indicating suppressed structural destruction (Fig. 1d). As displayed in Fig. 1e and 1f, there is a remarkable increase of capacity retention from 59.2% to 96.4% after LiBOB addition. Besides, the rate performance of LLO in LiBOB-containing electrolyte is superior to that without LiBOB (Fig. S4). At 5 C, the capacity with LiBOB is 104.9 mAh g^{-1} , significantly exceeding that without LiBOB (39.6 mAh g^{-1}). The cycling performance was further tested at an elevated temperature of 55 °C. While the cell fails at the third cycle in baseline electrolyte, the 0.1 M LiBOB addition results in a stable capacity around 260 mAh g^{-1} at 0.5 C after 50 cycles (Fig. S5).

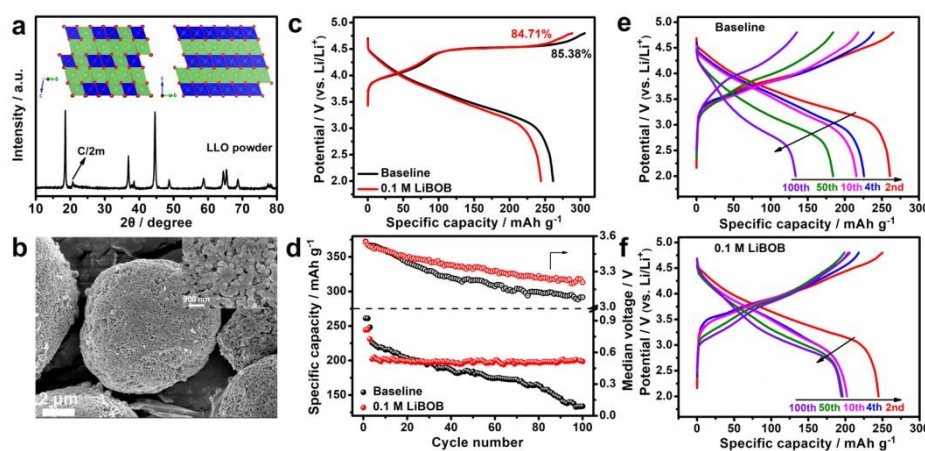


Figure 1 (a) XRD pattern and (b) SEM image of prepared LLO powder. Inset (a) shows schematic crystal structure. (c) The first charge and discharge curves at 0.1 C. (d) Cycling performance and midpoint voltage at 0.5 C of LLO cells in different electrolytes. Voltage profiles at selected cycles using (e) baseline and (f) LiBOB-containing electrolytes.

Fig. 2a and 2b show the differential capacity curves (dQ/dV) of LLO electrode in baseline and LiBOB-containing electrolytes, respectively. In the first cycle, there are two oxidation peaks at 4.0 and 4.5 V, which are the features of transition metal redox and activation of Li_2MnO_3 phase, respectively (Fig. S6).¹⁴ On cycling, the electrode in baseline electrolyte exhibits greater polarization with the anodic peak at 4.0 V shifting to higher voltage. In the presence of 0.1 M LiBOB, this peak shows a smaller offset, implying suppression of electrode polarization. Interestingly, an anodic peak appears at around 3.3 V after the addition of LiBOB, which is associated with the manganese redox and further reveals the contribution of LiBOB to long-term structural stability.³³ Fig. 2c and 2d display the discharge capacities above and below 3.5 V on cycling. The discharge capacities above and below 3.5 V are attributed to the redox of Ni and Mn species, respectively.^{14,34} The electrode suffers from severe capacity decay in baseline electrolyte. On the contrary, the deterioration above 3.5 V is largely mitigated while the capacity below 3.5 V gradually increases upon cycling with LiBOB (Fig. 2d). Surprisingly, the decreased capacity at the high voltage range can be compensated by the increased capacity at low voltages, maintaining a stabilized capacity of

200 mAh g^{-1} for 100 cycles. Thus, the addition of LiBOB evidently improves the cycling stability of LLO cathode.

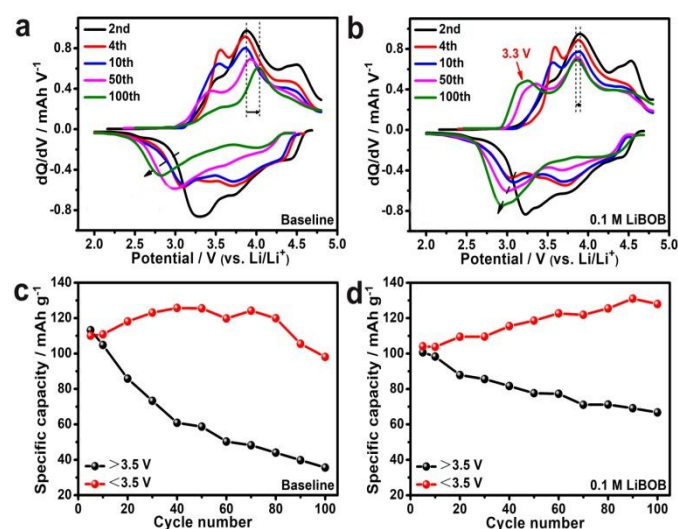


Figure 2: The corresponding dQ/dV curves (a) in baseline and (b) 0.1 M LiBOB electrolytes. Discharge capacity above and below 3.5 V of LLO cathode in electrolytes (c) without and (d) with 0.1 M LiBOB additive.

Electrochemical impedance spectroscopies (EIS) were recorded to analyse the interfacial kinetics of LLO in different electrolytes (Fig. 3). Each Nyquist plot roughly consists of a semicircle and a line region, which correspond to the charge-transfer impedance (R_{ct}) and the impedance of Li-ion diffusion, respectively. The presence of a suppressed semicircle may relate to the frequency overlapping of SEI impedance and R_{ct} , which together represents the total interface resistance.²¹ As shown in Fig. 3a, the interfacial impedance of battery in baseline electrolyte decreases firstly within 10 cycles and increases in subsequent cycles. Thus, the excessive

decomposition of electrolyte on the cathode surface on cycling could gradually destabilize the interface. In comparison, the cell with the electrolyte containing 0.1 M LiBOB experiences continuous reduction of resistance. Compared to the case with baseline electrolyte, the higher impedance with LiBOB addition at initial cycles could be a feasible attribution to the lower capacity of LLO (Fig. 1d). The results suggest that LiBOB could prevent the deterioration of the interface between LLO cathode and electrolyte and contribute to maintain facile interfacial charge transfer.

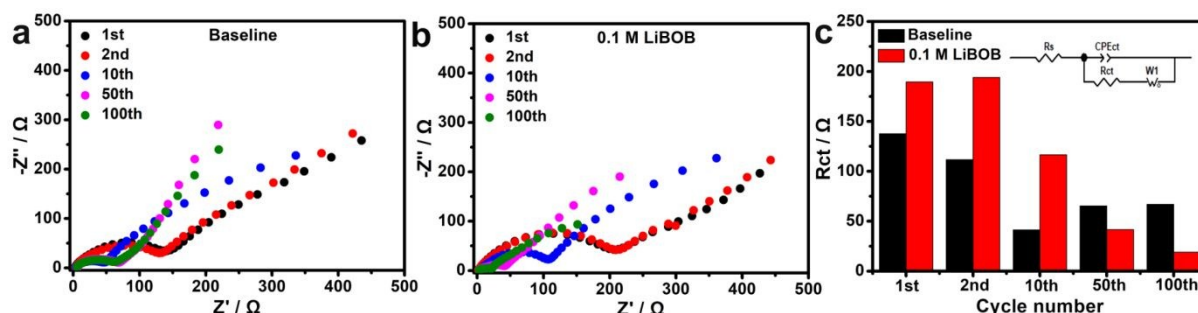


Figure 3: EIS spectra at different cycles of the cells with (a) baseline and (b) 0.1 M LiBOB-containing electrolytes. (c) Corresponding fitted results with the equivalent circuit models (inset).

The benefit of LiBOB in enhancing the performance of LLO motivated us to further investigate the underline mechanism. SEM images of the pristine and cycled cathodes (Fig. S7) show that micro-cracks and deposited by-products form on the particle surface after cycling in baseline electrolyte whereas the electrode in electrolyte with 0.1 M LiBOB features well-preserved spherical structure. Furthermore, from TEM imaging on the pristine LLO (Fig. 4a, 4d), clean and smooth particle surface can be clearly visualized along with (003) crystal planes featuring layered structure. After cycling in LiBOB-containing electrolyte, the layered structure of LLO is well-maintained with interlayer spacing of 0.476 nm, and a uniform film is observed on the primary particles (Fig. 4c, 4f). This CEI layer would ensure remarkable promotion of cycling performance. In contrast, there is no distinguishable surface film on the cycled cathode in baseline electrolyte (Fig. 4b). The LLO particle incorporates (004) and (311) facets of spinel phase with lattice spacing of 0.200 and 0.247 nm (Fig. 4e), suggesting phase transition.³⁵

Fig. S8 shows the XRD patterns of fresh and cycled cathodes in different electrolytes. The characteristic peaks of cathode in baseline electrolyte shift to lower angle than that with 0.1 M LiBOB additive, further evidencing protection of the layered structure by LiBOB. To confirm the inhibition effect of LiBOB on the transition metal (TM) dissolution, EDS and ICP spectroscopies were collected on cycled LLO electrode. As determined from EDS (Fig. S9), the proportion of Mn:Co:Ni (2.5:0.87:1) in the cathode cycled in baseline electrolyte apparently differs from that of the pristine LLO (4:1:1), whereas the electrode cycled with LiBOB additive exhibits a close ratio (3.95:1.01:1). The ICP measurement also reveals less deposition of transition metal elements on the anode extracted from the cell with LiBOB (Fig. 4g). These results

suggest that the generated CEI layer could be responsible for the structural and compositional integrity of LLO.

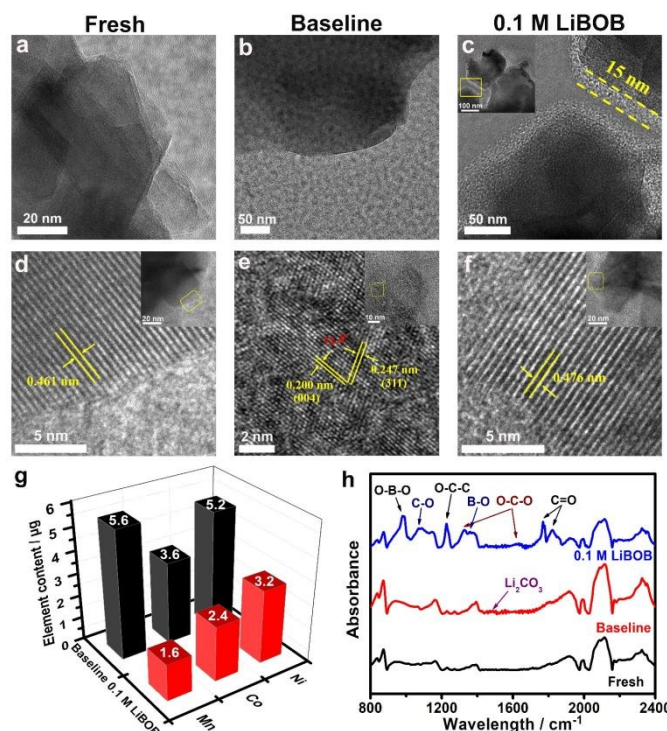


Figure 4: (a-c) Low-magnification and (d-f) high-resolution TEM images of (a, d) fresh cathode and the cathodes after 100 cycles (b, e) using baseline and (c, f) LiBOB-containing electrolytes. Insets in (c, f) show the location for HRTEM imaging. (g) ICP measurement of transition metals deposited on lithium metal anode after 100 cycles with and without LiBOB additive. (h) FTIR spectra of fresh and LLO cathodes after cycling in baseline and 0.1 M LiBOB-containing electrolytes.

FTIR spectra of LLO were collected to probe the dominant components in the CEI film. As displayed in Fig. 4h, additional absorption signals detected on the cathode surface in 0.1 M LiBOB electrolyte are marked with asterisks, which differ the infrared spectrum of DMC solvent (Fig. S10b). The existence of two characteristic peaks at 1823 and 1778 cm^{-1} is characteristic of the carbonyl group in dissolved BOB anion.³⁶ However, BOB⁻ undergoes variable oxidation processes on cycling, and the DFT calculation reveals that semi-BOB species (Fig. 5) obtained from BOB⁻ decomposition owns similar vibration frequency of carbonyl (Table S1). From theoretical calculations, the peak at 1226 cm^{-1} may arise from the O-C-C asymmetric stretching model and the absorption peak at 987 cm^{-1} can be attributed to the O-B-O vibration model, both of which emerge in the semi-BOB structure and BOB anion.³⁶ Therefore, unreacted BOB⁻ or semi-BOB constituents may exist in the CEI film. As compared with the FTIR spectrum of lithium oxalate (Fig. S10a), the absorption at 1650 and 1331 cm^{-1} can be assigned to the O-C-O moiety in oxalates,³⁶ which stems from the decomposition of LiBOB.³⁷ The 1358 and 1074 cm^{-1} signals are assigned to the B-O asymmetry stretching modes and the C-O symmetric stretching vibration, respectively,³⁶ as referenced to the spectrum of trimethylborate (Fig. S10a). Nevertheless, the characteristic peaks for Li_2CO_3 at 1412 and 857 cm^{-1} were not detected on the surface of cycled cathode with 0.1 M LiBOB addition, confirming limited parasitic reactions between solvents and oxygen species.¹⁴

To evaluate the applicability of LiBOB, the electrode performances of two common cathode materials, namely $\text{LiNi}_{0.8}\text{Co}_{0.1}\text{Mn}_{0.1}\text{O}_2$ (NCM811) and $\text{LiNi}_{0.5}\text{Mn}_{1.5}\text{O}_4$ (LNMO), were tested in baseline and 0.1 M LiBOB-containing electrolytes. The cyclability of both NCM811 and LNMO with LiBOB evidently outperform that in baseline electrolyte (Fig. S11). The FTIR analysis of the two cycled cathodes was carried out to gain insight into the effect of electrolyte additive. As shown in Fig. S12, only two characteristic peaks are observed at 1810 and 1780 cm^{-1} after adding LiBOB for both LNMO and NCM811 electrodes. The absorption corresponds to the carbonyl group in BOB anion and differs from the signals on LLO (Fig. 4h), indicating a dissimilar oxidation path of LiBOB to the case of Li-rich cathode materials that involve anionic redox.

The composition of the CEI film was further analysed by XPS measurement, as shown in Figure S13. The characteristic peaks of B-O (192.5 eV)²¹ can be recognized in the B 1s spectra for the cathode with 0.1 M LiBOB, confirming the participation of LiBOB in the formation of CEI layer. In the C 1s spectrum, the increased C=O signal can be attributed to the oxalate species generated from the oxidation of LiBOB,³⁴ agreeing well with the FTIR results. Additionally, the lower proportion of LiF indicates less decomposition of LiPF_6 after the addition of LiBOB.^{18,21} Combining all these characterization results uncovers that the composition of formed CEI film on LLO in 0.1 M LiBOB electrolyte is dominated by partially unreacted BOB anion, semi-BOB structure, alkyl borates, oxalates and a small amount of lithium fluoride.

As the presence of active oxygen (denoted as [O]) in LLO is previously proposed to affect the oxidation process of

electrolyte,^{21,22} we investigated the LiBOB-[O] interaction to further clear the mechanism of LiBOB by DFT calculation. The optimized structures of EC, EMC and BOB⁻ are shown in Fig. S14. There are two proposed pathways for LiBOB oxidation.^{38,39} In the first route, BOB⁻ eliminates one CO_2 molecule to form a boron-containing radical, followed by two radicals binding to generate 2B structure (Fig. 5a process I).³⁸ In the other route (Fig. 5a process II-III), two CO_2 molecules are extracted from BOB⁻ anion to generate 1B, which bonds with EC to form 1B-EC complex.³⁹ The calculated Gibbs free energy (ΔG) of path II is lower than of path I (4.37 vs. 4.74 eV), suggesting easier decomposition of BOB⁻ to get 1B. However, as mentioned above, the active oxygen is related to the path of electrolyte oxidation. Comparing the ΔG of the reactions of oxygen radicals with EC, EMC and LiBOB (Fig. S15), boron-containing compound would trap the oxygen prior to solvents due to the lowest ΔG of 1B-O complexes (Fig. 5b and Fig. S15). The energy profiles suggest that LiBOB has a tendency to oxidize through paths from II to IV shown in Fig. 5a.

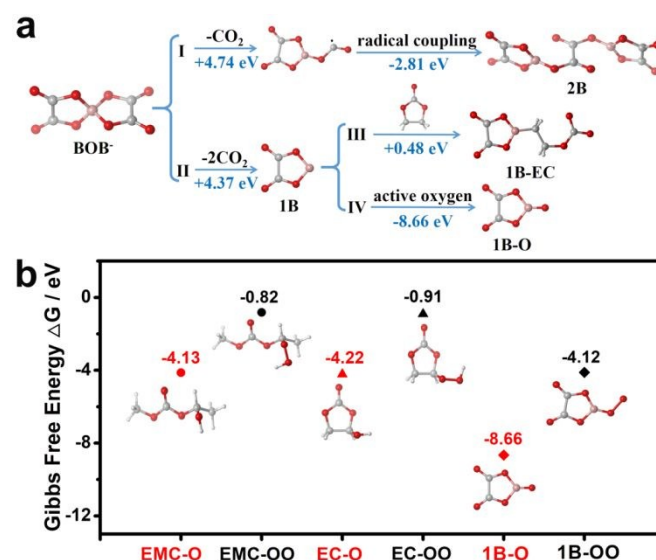


Figure 5: (a) The possible oxidative decomposition processes of LiBOB additive. (b) Optimized structures and Gibbs free energies of EC and EMC solvents and LiBOB additive with active oxygen radicals ($\text{O}_2^{\cdot-}/\text{O}^{\cdot-}$).

Based on experimental and computational results, the contribution of LiBOB to the enhanced stability of LLO cathode is schematically illustrated in Fig. 6. In baseline electrolyte, the generated $\text{O}_2^{\cdot-}/\text{O}^{\cdot-}$ species via Li_2MnO_3 activation could gradually accelerate the oxidation of carbonate-based solvents to produce CO_2 , CO and organic by-products, incurring continuous consumption of electrolyte. An uneven CEI film is gradually formed on the cathode surface as the cycling proceeds. This CEI film is invalid to protect the surface structure from the continuous attack by acidic components (e.g., HF). With 0.1 M LiBOB additive, oxygen radicals tend to combine with 1B to form 1B-O anion, slowing down the electrolyte oxidation. Lithium difluoro(oxalate) borate (LiDFOB) is proposed to react with Mn ions of LiMn_2O_4 to hinder cation dissolution.⁴⁰ Similarly, the 1B-O anion remains chemisorbed with transition metal ion on cathode surface

(process I). The 1B-O anion may undergo C-C cleavage to form radicals (process II).⁴¹ Like in the case of LiDFOB that two adjacent F₂BOCO radicals interact with each other,⁴² a couple

of radicals generated in process II would recombine to form a crab-like structure (process III), which occurs as a stable component of the protective CEI layer on LLO cathode.

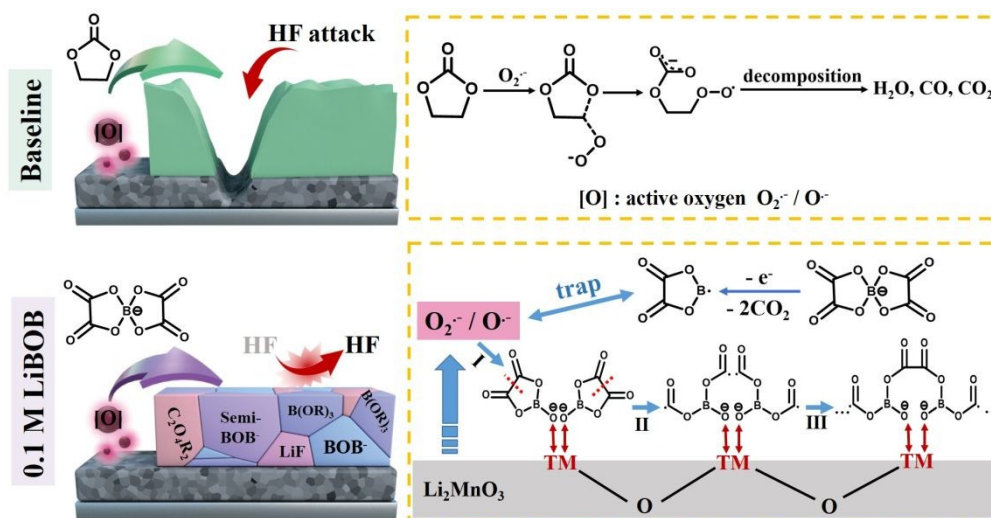


Figure 6: Schematic illustration showing the protective effect and proposed reactions associated with LLO cathode in electrolyte with LiBOB additive (TM stands for transition metal ions such as Mn, Ni, and Co).

Conclusion

In summary, LiBOB has been introduced as an effective interphase-forming additive to stabilize the LLO cathode with composition of $\text{Li}_{1.2}\text{Mn}_{0.54}\text{Co}_{0.13}\text{Ni}_{0.13}\text{O}_2$. With a formulated 0.1 M LiBOB, the capacity retention of attains 96.4% at 0.5 C after 100 cycles, significantly outperforming that of the counterpart without any additive. A combination of electrochemical analysis, EDS, ICP, SEM and TEM reveals that the remarkable cycling performance could be attributed to the formation of protective surface layer that suppresses irreversible side reactions and helps to maintain compositional and structural integrity. Theoretical calculations and ex-situ surface characterizations suggest preferential combination of LiBOB and oxygen radicals to form boron-enriched layer containing oxalate, lithium fluoride and alkyl borate constituents. The results would enlighten the design of boron-based electrolyte additive for cathode materials that require high charging voltage to release the deliverable capacities.

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

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